

INTELLIGENT DECISION ENGINE & DIGITAL TWIN

Visual Risk AI

AI-Based Visual Risk & Compliance Intelligence System

A deterministic mathematical simulation engine, autonomous circadian schedule planner, and multi-modal conversational copilot backed by MongoDB persistence.

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Platform: React 19 + FastAPI + MongoDB

Executive Summary: The Mission of Visual Risk AI

THE PROBLEM

- Fragmented Life Telemetry

Health trackers, bank apps, and calendars operate in isolated silos without holistic risk awareness.

- Hidden Compounding Friction

A minor daily sleep reduction or impulse purchase cascades into 5-year compounding milestone delays.

- Static Unresponsive Tools

Traditional planners require tedious manual scheduling and fail to compensate for biological sleep debt or burnout.

- Lack of Stochastic Modeling

Spreadsheets assume linear outcomes, failing to model real-world market volatility and probabilistic risk.

THE VISUAL RISK AI SOLUTION

- Unified Digital Twin State

Real-time mathematical aggregation of biometrics, study sprints, cashflows, and long-term milestones.

- 500-Run Monte Carlo Engine

Geometric Brownian Motion simulation providing probabilistic percentile bounds (P10, P50, P90).

- Autonomous Circadian Planner

Auto-schedules daily focus blocks aligned with peak cortisol curves (08:30-11:30 AM) and sleep debt.

- 4-Stage Agentic Copilot

Conversational AI that computes mathematical tradeoffs and yields 1-click executable actions into MongoDB.

End-to-End Multi-Tier System Architecture

Tier 1: Client Layer

React 19 + TypeScript + Vite

- Interactive Digital Twin Dashboard
- Visual Risk Copilot (/chat) with Voice STT
- Decision Sandbox with Dual Sliders
- Autonomous Planner with Timeline Visuals
- Habit Analytics with Correlation Graphs

Tier 2: Backend Gateway

FastAPI + Motor + Beanie ODM

- Asynchronous High-Throughput REST API
- Argon2id/Bcrypt + JWT & Rotating Cookies
- Token Family Theft & Replay Protection
- Unified Multi-Role Authentication
- Structured Pydantic v2 Request Validation

Tier 3: Intelligence Engine

Groq LLMs + NumPy Simulation

- 4-Stage Agentic Reasoning Pipeline
- 500-Trial Stochastic Monte Carlo Engine
- Circadian Cortisol & Alertness Modeler
- Dynamic Markdown Schedule Table Parser
- Groq 120B/20B/27B with Rule Fallback

Tier 4: Persistence Layer

MongoDB + Local Snapshot Store

- Beanie Document ODM Collections
- Embedded Messages for Atomic Dialogue
- Automatic Snapshotting on All Mutations
- data/mongodb_persistence.json Store
- Zero-Loss Session Recovery Across Logins

Target Personas & Dynamic Telemetry Profiles

COLLEGE STUDENT — Age 20 | Exam Readiness

Coursework mastery, study sprint planning, budget control, and consistent sleep cutoff.

WORKING PROFESSIONAL — Age 29 | Senior Promotion

Deep work focus sprints, career velocity, liquid portfolio growth, and burnout prevention.

FREELANCER / CREATOR — Age 26 | Revenue Smoothing

Irregular cashflow buffering, client milestone time-boxing, and vitality recovery.

ENTREPRENEUR / FOUNDER — Age 32 | Venture Scaling

Cash runway optimization, investor pitch sprint scheduling, and high-stress resilience.

ACTIVE RETIREE — Age 62 | Capital Preservation

Stochastic longevity modeling, health maintenance, estate planning, and lifestyle vitality.

Decision Sandbox: Real-Time What-If Tradeoff Engine

DUAL SCENARIO STRESS-TESTING

- Scenario A vs Scenario B

Compares live baseline against simulated adjustments across Sleep, Study, and Monthly Savings.

- Non-Linear Vitality Modeling

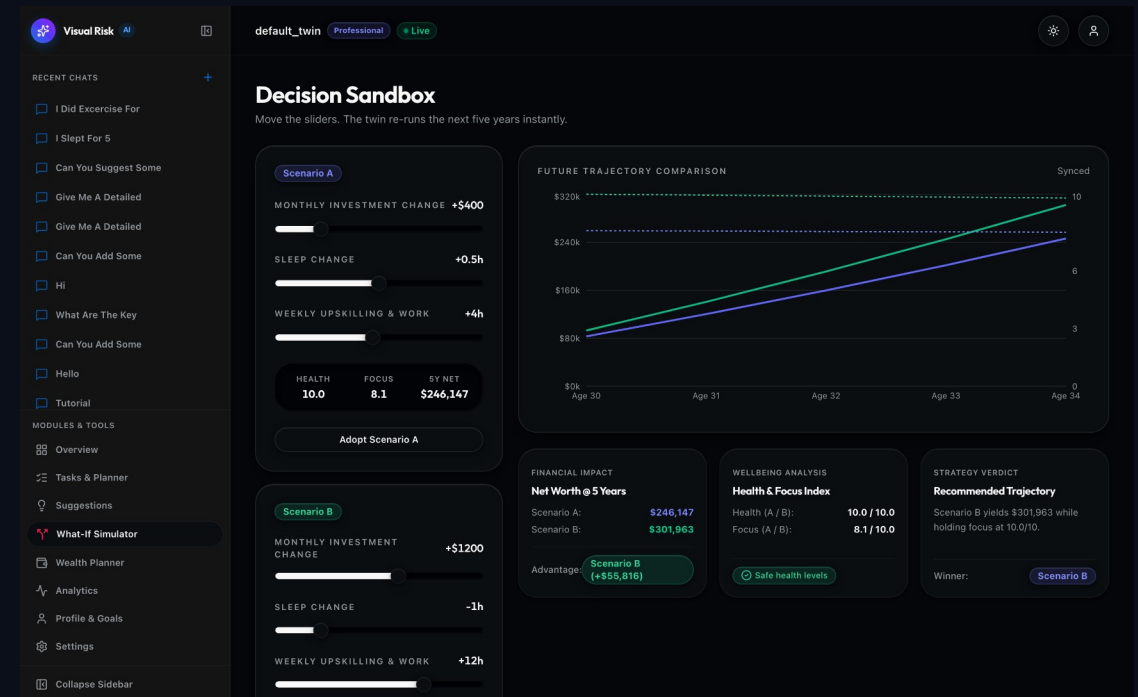
Reductions in sleep below baseline invoke an exponential penalty on daytime cognitive focus.

- Compound Capital Opportunity Cost

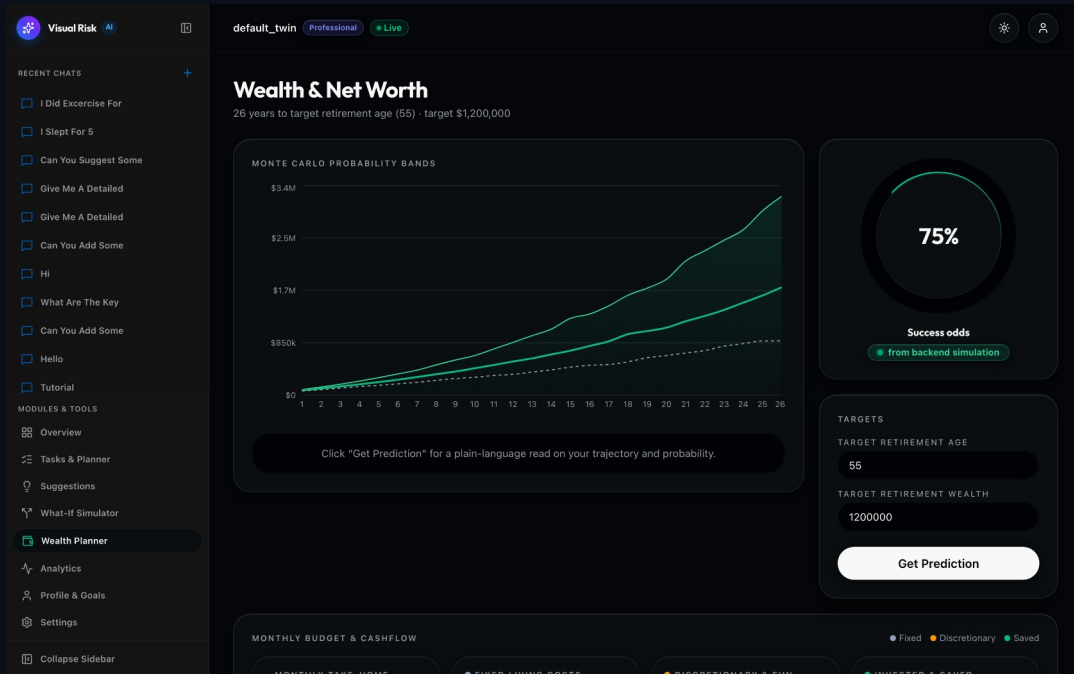
Models monthly savings deltas compounded at 8% CAGR over 1, 3, and 5-year investment horizons.

- 1-Click Scenario Preset Adoption

Seamlessly commits validated What-If scenario presets into MongoDB UserDoc profile settings.



Stochastic Wealth Planner: 500-Run Monte Carlo Simulation



GEOMETRIC BROWNIAN MOTION (GBM)

- 500 Stochastic Simulation Paths

Simulates annual portfolio drift and market volatility across 500 distinct trials to target retirement age.

- Percentile Bounds (P10, P50, P90)

Computes realistic worst-case floor (10th percentile), median trajectory (50th), and optimistic upside (90th).

- Goal Attainment Probability

Calculates exact percentage confidence score of exceeding net worth target by goal age.

- Deterministic vs Stochastic Comparison

Exposes variance risk between static linear savings models and market volatility reality.

Autonomous Circadian Daily Planner & Governance

CIRCADIAN TIME-BLOCKING ENGINE

- Peak Cortisol Alignment (08:30-11:30)

Automatically places deep work and difficult academic coursework into prime cognitive focus windows.

- Sleep Debt Compensation

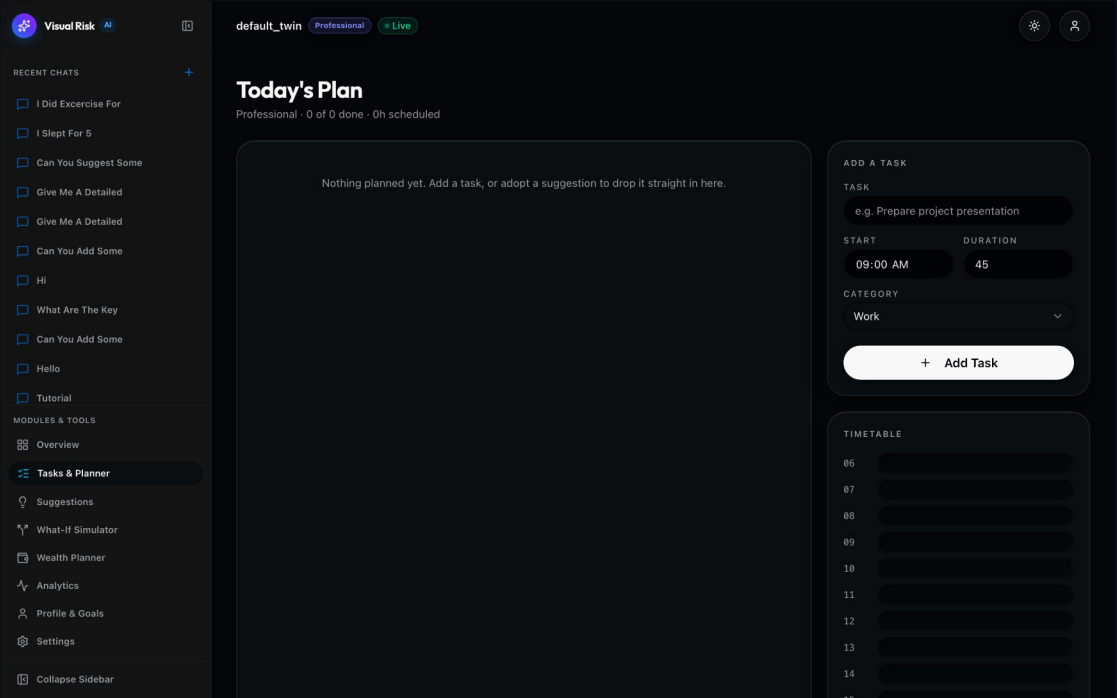
Dynamically evaluates 7-day rolling sleep debt; lightens sprint density on sleep-deprived mornings to prevent burnout.

- 3-Tier Autonomy Governance

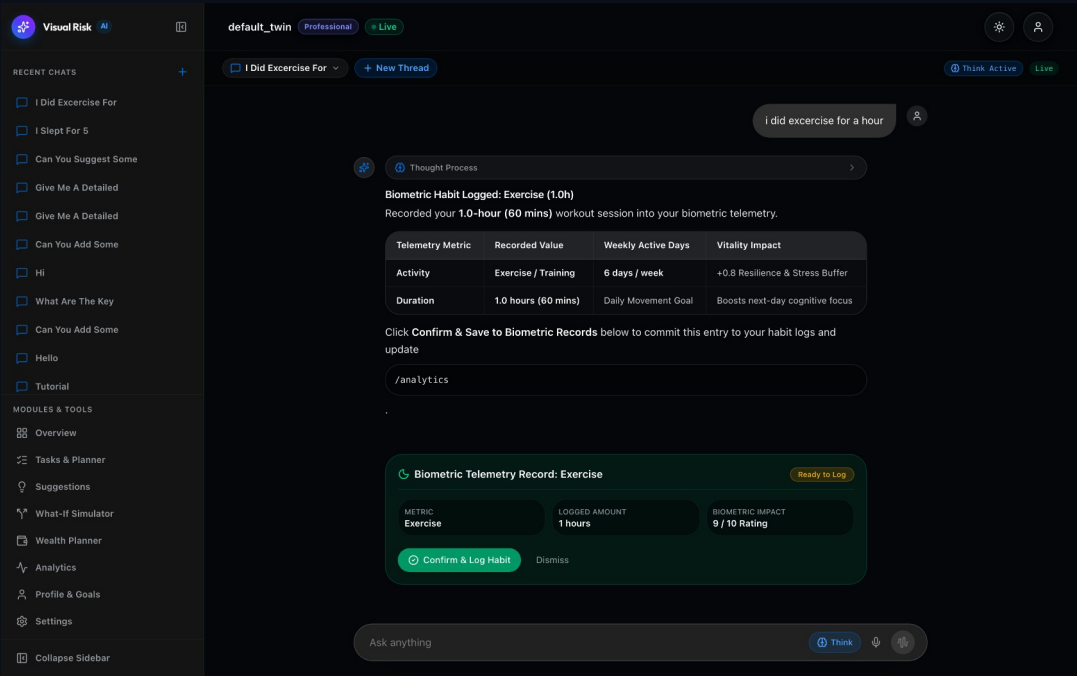
Supervised (Manual approval required), Semi-Autonomous (Auto-executes safe tasks), Full-Autonomous (Proactive scheduling).

- Morning AI Briefings

Synthesizes contextual daily briefings with scheduled focus blocks and biometric readiness scores.



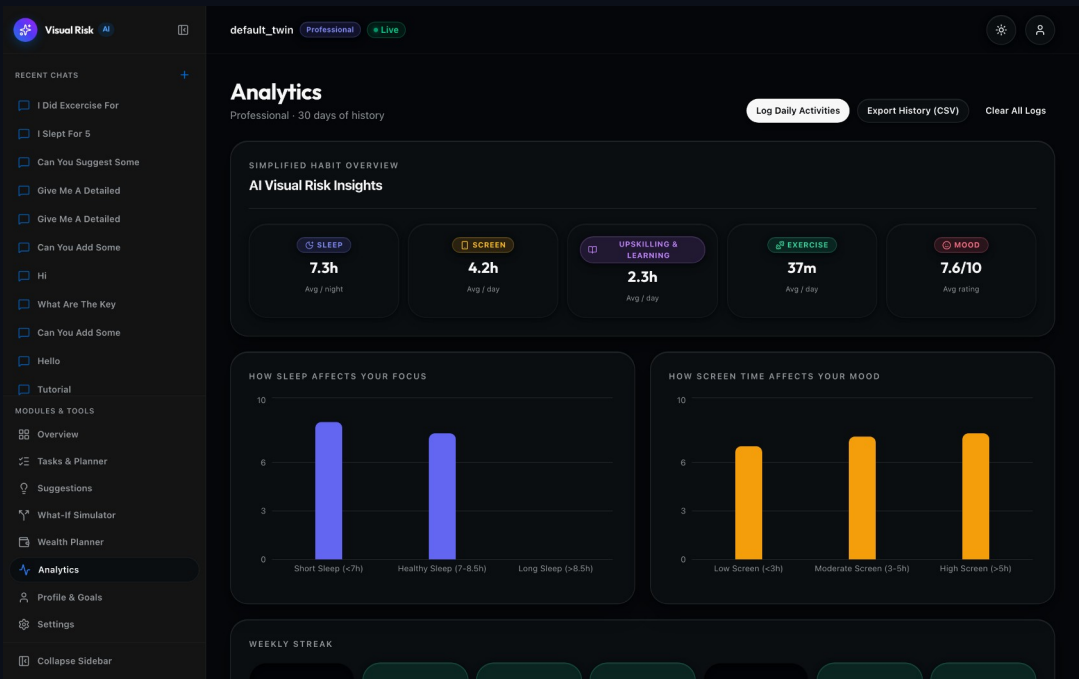
Visual Risk Copilot: 4-Stage Agentic Reasoning



4-STAGE REASONING PIPELINE

- **Step 1: Goal Definition**
Decomposes user inquiry into explicit optimization targets and friction domains.
- **Step 2: Telemetry Gathering**
Queries MongoDB for live habit records, study logs, cashflow surplus, and goal progress.
- **Step 3: Multi-Criteria Analysis**
Evaluates biological elasticity, circadian cortisol curves, and stochastic forecasts.
- **Step 4: Strategic Plan & Action Card**
Generates reasoning tree (<think>), markdown table, and 1-click executable action proposal.

Habit Analytics & Predictive Correlation Modeling



ANALYTICS CAPABILITIES

- **Grouped Correlation Analysis**
Models cross-impact between Sleep Duration, Screen Time, Exercise Frequency, and Focus Scores.
- **Automated 12:00 PM Reflection**
Daily noon cron that synthesizes 24-hour retrospective insights cached in MongoDB AppCache.
- **Vitality & Focus Elasticity**
Quantifies marginal focus gains per additional hour of sleep or reduction in late-night screen time.
- **Habit Consistency Tracking**
30-day historical time series visualization with dynamic trend lines and standard deviation bands.

Academic & Coursework Productivity Intelligence

7-Day Pomodoro Sprints

AI Study Plan Synthesis

- Synthesizes customized 7-day revision timetables based on active enrolled courses.
- Allocates 45-min Pomodoro sprints with 15-min cognitive recovery intervals.
- Prioritizes high-weight and under-studied subjects automatically.

Spaced Repetition Engine

Memory Decay Mitigation

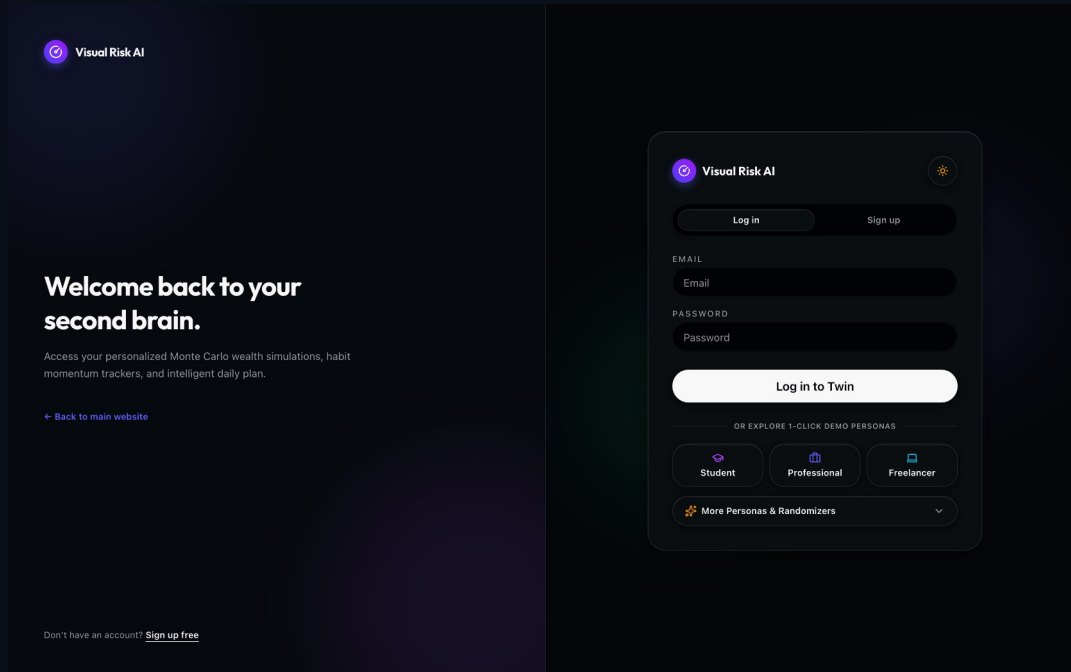
- Applies Ebbinghaus forgetting curve intervals to reinforce key syllabus concepts.
- Tracks focus ratings per study session to identify cognitive fatigue thresholds.
- Logs coursework hours into MongoDB StudyRecordDoc collections.

Exam Score Forecaster

Probabilistic Grade Modeling

- Predicts expected exam performance and variance based on cumulative study hours.
- Visualizes readiness radar across core modules and topics.
- Provides 1-click study sprint scheduling into the Daily Planner.

Multi-Session Authentication & Cryptographic Security



CRYPTOGRAPHIC PROTOCOLS

- Stateless JWT + Rotating Refresh Cookies

Short-lived access tokens (15 mins) paired with HttpOnly SameSite rotating refresh cookies (7 days).

- Token Family Theft Detection

Maintains token family trees in MongoDB. If a revoked token is re-submitted, the entire family is revoked immediately.

- Argon2id & Bcrypt Password Hashing

Cryptographically secure password hashing with cost factor 12 and salt isolation.

- Enumeration-Safe Generic Auth

Consistent generic error responses prevent account enumeration on login and password recovery.

- Single-Use Random Recovery Tokens

Cryptographically generated 48-byte URL-safe tokens with strict TTL expiration.

MongoDB Document Architecture & Disk Persistence

BEANIE DOCUMENT COLLECTIONS

- UserDoc: Profiles, targets, net worth, tasks_json, autonomy_mode, and slider presets.
- ChatSessionDoc: Conversational threads with embedded ChatMessageDoc lists for atomic fetch.
- HabitRecordDoc: Biometric sleep, screen time, exercise, and daily vitality logs.
- StudyRecordDoc: Academic coursework, focus ratings, and subject sprint logs.
- FinancialRecordDoc: Income and expense transactions categorized for cashflow analysis.
- UserSuggestionDoc: AI-generated recommendations with adoption states and impact scores.

LOCAL SNAPSHOT PERSISTENCE ENGINE

- Synchronous Mutation Sync

Every database write operation automatically triggers `save_persistence_snapshot()`.

- Atomic Temporary File Swap

Writes to `data/mongodb_persistence.json.tmp` before atomic OS replace to eliminate corruption.

- Lossless Session Recovery

Tutorial threads, user custom chats, and action states are completely preserved across logins and reboots.

- Embedded Engine Fallback

Allows 100% offline standalone operation with zero external cloud dependencies.

100% PASS RATE & RIGOROUS VERIFICATION

Quality Assurance, Testing & Automated Verification

21 / 21 Tests Passing

Pytest Suite (100%)

Comprehensive test coverage spanning mathematical simulation formulas, Pydantic schemas, auto-planner algorithms, and multi-turn table parsers.

0 Build Errors

TypeScript + Vite

Clean frontend compilation across React 19, Radix UI primitives, Lucide icons, and Tailwind styling with zero type warnings.

Lossless Rehydration

End-to-End Verified

Full lifecycle verification across user registration, message appending, disk snapshotting, in-memory purging, and persistent rehydration.

Multi-Persona Integrity

5 Personas Calibrated

Realistic 30-day telemetry baselines and tailored suggestions verified for Student, Professional, Freelancer, Founder, and Retiree.

Key Innovation Highlights & Future Roadmap

CORE INNOVATION HIGHLIGHTS

- Deterministic & Stochastic Fusion

Combines linear biological feedback modeling with 500-run Monte Carlo stochastic uncertainty.

- Autonomous Circadian Orchestration

Transitions productivity from static calendars to intelligent, sleep-debt-aware schedule synthesis.

- Agentic Reasoning with 1-Click Execution

Eliminates conversational dead-ends by turning AI reasoning directly into database mutations.

- Zero-Data-Loss Embedded Durability

Delivers enterprise-grade reliability with JSON disk snapshots and MongoDB Beanie ODM.

FUTURE PRODUCT ROADMAP

- Wearable Biometric Sensor Sync

Direct real-time Bluetooth and API synchronization with Apple Watch, Oura, and Garmin devices.

- Multi-Agent Collaborative Simulations

Simulating multi-party financial negotiations, corporate risk modeling, and team sprints.

- On-Device Local Neural Inference

WebGPU-accelerated local SLMs for zero-latency private biometric analysis.

- Automated Banking & Open Banking Ingestion

Plaid API integration for real-time transaction reconciliation and spending categorization.

THANK YOU

Visual Risk AI

Empowering Intelligent Decisions Through Mathematical Simulation & Circadian AI

Project Repository: [AI-Based-Visual-Risk-and-Compliance-Intelligence-System](#)

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Core Stack: React 19, FastAPI, Motor/Beanie, MongoDB, NumPy, Groq API